

Layer changes at bends

Real detailed routes switch between M1 and M2 at vias

The idea

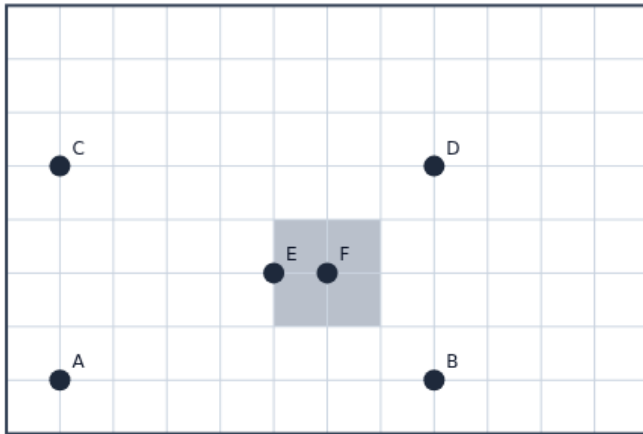
- Given start and goal
- `L_route_layers` returns segment dicts with x y layer and optional via flag
- A path from one comma one to five comma four in HV includes at least one via

L-bend via

VIA ASSIGNMENT

L-bend via

Step 1 / 5 · bend · module02-07-via-assignment



Explanation

Layered L places a via at the corner to switch M1 → M2.

- HV / VH

Orange dot

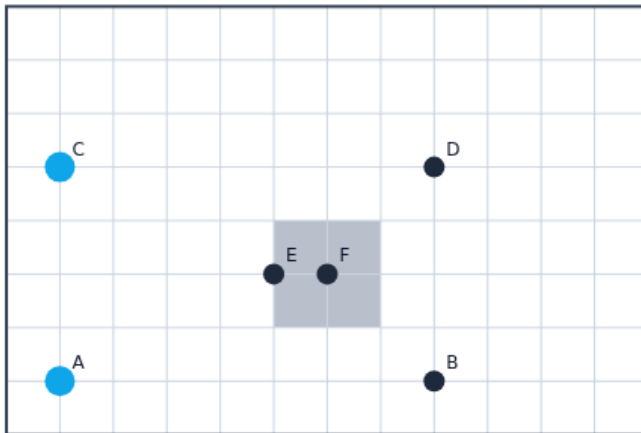
L-bend via

A-C example

VIA ASSIGNMENT

A-C example

Step 2 / 5 · ab · module02-07-via-assignment



Explanation

Spread A-C layered path uses M1 then M2 with via.

- Same pins

Via at bend

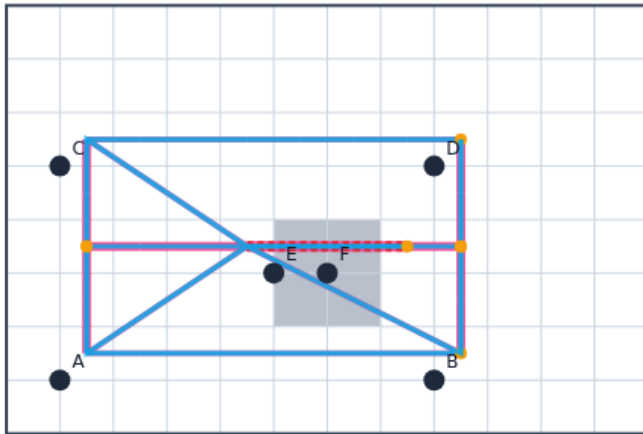
A-C example

Canvas vias

VIA ASSIGNMENT

Canvas vias

Step 3 / 5 · draw · module02-07-via-assignment



Explanation

Walkthrough marks via points on the grid.

- Layer discipline

Marked vias

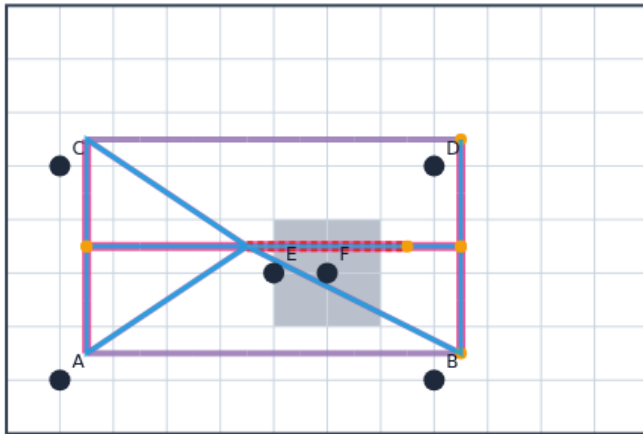
Canvas vias

Multi-pin

VIA ASSIGNMENT

Multi-pin

Step 4 / 5 · multi · module02-07-via-assignment



Explanation

Four-pin net concatenates L legs—multiple vias possible.

- Star center

Four legs

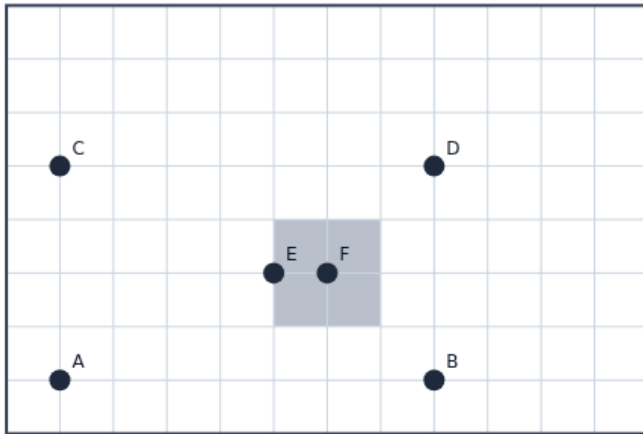
Multi-pin

Spacing next

VIA ASSIGNMENT

Spacing next

Step 5 / 5 · drc · module02-07-via-assignment



Explanation

Vias do not replace DRC spacing checks.

- Lite checker

DRC 1ab

Spacing next

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Via assignment

INTERACTIVE TOOL

Via assignment

Layered L routes use vias at bends to switch M1 ↔ M2.

Your job: layered routes place vias at L-bends. Route layered and inspect metrics.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] A-B has via on L

Pick a challenge and click **Start**. Move cells, route, then **Check**.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓ Route Lee Route layered Route A*

Rip-up detailed

Graph

Reference / metrics

selected: A @ grid (1,1)
routes: (none — click Route)
overflow total: 0.00 · max: 0.00 · count: 0
track capacity: 2 · mode: astar
HPWL: 34.0
net0 vias: 0

Browser lab starter

learn_routing — 07 via assignment

Implement track

- Implement ``l_route_layers(a, b, prefer)``
- Route net A-B on `tiny_dr` and print segments with via flags
- Match the browser overlay

Pitfalls

- Skipping the via marker at the bend
- Putting horizontal motion on M2 or vertical on M1
- Forgetting the final segment must land on the goal coordinates

Your turn

- Ship Track A layer routes with vias
- Next: DRC spacing lite on parallel tracks